Advances made in fuel cells and hydrogen-producing electrolysers

MIT chemists have made a groundbreaking discovery in the vital chemical process crucial for energy technologies like fuel cells and hydrogen-producing electrolysers. They have elucidated the proton-coupled electron transfers at an electrode surface, a critical step in generating electric currents. The team designed electrode surfaces using graphene sheets linked to organic compounds, exploring proton movement. The pH of the solution was found to affect proton transfer rates, with faster rates at extreme pH levels. Surprisingly, equilibrium occurred at pH 10, challenging existing models. The researchers are now delving into how various electrolyte ions influence the speed of proton-coupled electron flow for potential advancements in fuel cells and batteries.

A fuel cell that uses soil microbes to power underground sensors

Researchers at Northwestern University have developed a fuel cell that uses soil microbes for power, similar in size to a paperback book, offering an energy solution for underground sensors in precision agriculture and infrastructure. This technology provides a renewable alternative to conventional batteries and can power soil moisture sensors and track animal movements without the environmental issues of traditional battery production. It includes a small antenna for wireless data transmission, enhancing its utility. The fuel cell performs well in various soil conditions, lasting longer than similar technologies by 120%. After two years of development and testing in an outdoor garden, the team's design combines unique geometry with a carbon-felt anode and a metal cathode for efficient operation in both dry and wet conditions. This work leads to sustainable, conflict-free computing solutions, aiming for a fully biodegradable version.

The fuel cell in the lab (Credit: https://news.northwestern.edu)

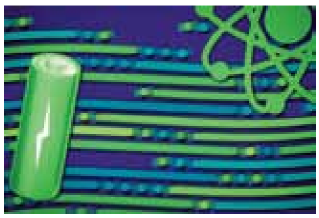
New AI wireless tech might surpass 5G speed, reliability, and coverage

Researchers at UBC Okanagan have developed a wireless communication framework that aims to surpass 5G in terms of speed, reliability, and coverage. They are working on an integrated system to enable smooth connectivity between devices, users, and the environment. Dr Chaaban is using artificial intelligence to tackle the complexities of future networks. His team uses AI techniques, like transformer-masked autoencoders, to improve data transmission by smartly handling content delivery and recovery. Their work focuses on fulfilling the needs for extensive connectivity, low latency, and high reliability, emphasising energy efficiency and cost-effectiveness. Dr Chaaban's project uses AI to navigate the challenges of emerging technologies, such as virtual reality, aiming to create adaptable, efficient, and secure wireless communication systems.

Next-generation mobile networks are expected to outperform 5G on many fronts (Source: https://news.ok.ubc.ca)

Modified electrode surfaces improve efficiencies of fuel cells and batteries

MIT chemists have detailed the process of proton-coupled electron transfers at electrode surfaces, which is crucial for energy technologies like fuel cells and electrolysis producing hydrogen gas. They developed a method to modify electrode surfaces with graphene sheets connected to organic compounds, enhancing proton attraction and facilitating electron flow into graphene. This approach enables the measurement of proton transfer rates at equilibrium, showing that the reaction's speed changes with the solution's pH, being quicker in acidic or basic conditions. Their findings contradict previous models by demonstrating that reaction rates are equal at a pH of 10, not neutral pH. This could improve the efficiency of fuel cells, batteries, and other energy technologies. The team is investigating how different ions influence proton-coupled electron flow speeds.



MIT chemists have mapped out how proton-coupled electron transfers happen at the surface of an electrode (Source: MIT News, iStock)

Automated deep learning system to help improve product design and brand management

Researchers at Carnegie Mellon University have developed BIGNet, an automated deep learning system to improve brand consistency across products, which could lead to